FLATNESS OF AN INVERSE LIMIT OF FLAT MODULES OVER THE QUOTIENTS BY POWERS OF AN IDEAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Tor against a quotient by a power of an ideal is an essentially constant pro-zero system. Lemma \ref{lemma-tor-strictly-pro-zero}, cited at proof lines 6498-6499. Makes the obstruction to commuting tensor products with the limit vanish.
- Mittag-Leffler style lemma identifying the limit of a cokernel with the cokernel of the limits. Homology, Lemma \ref{homology-lemma-apply-Mittag-Leffler-again}, cited at proof lines 6500 and 6519. Produces the isomorphism between the tensor product with the limit and the limit of the tensor products, and the injectivity of the limit of the maps.
- Flatness is detected by injectivity of tensoring with injective maps of modules. Algebra, Lemma \ref{algebra-lemma-flat}, cited at proof lines 6508-6509. Turns the second part of the proof into a criterion for flatness of the limit module.

Source reference. Original lemma 0912: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11430–11430. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0911: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11429–11429. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 070E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8953–8953. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 639–639. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A be a Noetherian ring. Let I be an ideal of A. Let (M_n) be an inverse system of A-modules such that*

(1) *M_n is a flat A/I^n-module,*
(2) *$M_{n+1} \to M_n$ is surjective.*

Then $M = \lim M_n$ is a flat A-module and $Q \otimes_A M = \lim Q \otimes_A M_n$ for every finite A-module Q.

Proof. We first show that $Q \otimes_A M = \lim Q \otimes_A M_n$ for every finite A-module Q. Choose a resolution $F_2 \to F_1 \to F_0 \to Q \to 0$ by finite free A-modules F_i. Then

$$F_2 \otimes_A M_n \to F_1 \otimes_A M_n \to F_0 \otimes_A M_n$$

is a chain complex whose homology in degree 0 is $Q \otimes_A M_n$ and whose homology in degree 1 is

$$\mathrm{Tor}_1^A(Q, M_n) = \mathrm{Tor}_1^A(Q, A/I^n) \otimes_{A/I^n} M_n$$

as M_n is flat over A/I^n. By Lemma 0911 we see that this system is essentially constant (with value 0). It follows from Homology, Lemma 070E that $\lim Q \otimes_A M_n = \mathrm{Coker}(\lim F_1 \otimes_A M_n \to \lim F_0 \otimes_A M_n)$. Since F_i is finite free this equals $\mathrm{Coker}(F_1 \otimes_A M \to F_0 \otimes_A M) = Q \otimes_A M$.

Next, let $Q \to Q'$ be an injective map of finite A-modules. We have to show that $Q \otimes_A M \to Q' \otimes_A M$ is injective (Algebra, Lemma 00HD). By the above we see

$$\mathrm{Ker}(Q \otimes_A M \to Q' \otimes_A M) = \mathrm{Ker}(\lim Q \otimes_A M_n \to \lim Q' \otimes_A M_n).$$

For each n we have an exact sequence

$$\mathrm{Tor}_1^A(Q', M_n) \to \mathrm{Tor}_1^A(Q'', M_n) \to Q \otimes_A M_n \to Q' \otimes_A M_n$$

where $Q'' = \mathrm{Coker}(Q \to Q')$. Above we have seen that the inverse systems of Tor's are essentially constant with value 0. It follows from Homology, Lemma 070E that the inverse limit of the right most maps is injective. □